

Daily Geometry Drill #5

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Angle at the centre · Cyclic quadrilaterals · Tangent $\perp$ radius · Power of a point · Touching circles: join the centres.

Attempt each question before checking answers. Time yourself. No calculators. Diagrams are not to scale.

Section A Rapid Recognition

(≤ 15 s each)

Angle at the centre, angles on the same arc, cyclic opposite angles, the alternate segment, and power of a point. Know them cold.

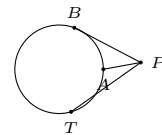
- In a circle with centre O , the angle at the circumference subtending arc AB is 20° . Write down the angle at the centre.
- AB is a diameter of a circle and P is any other point on the circle. Write down $\angle APB$.
- $ABCD$ is a cyclic quadrilateral with $\angle A = 70^\circ$ and $\angle B = 85^\circ$. Write down $\angle C$.
- The tangent at T to a circle makes an angle of 40° with the chord TA . By the alternate segment theorem, write down the angle in the alternate segment subtended by TA .
- Chords AB and CD of a circle intersect inside at P , with $PA = 2$, $PB = 6$ and $PC = 3$. Write down PD .
- A tangent from P touches the circle at T , and a secant through P cuts the circle at A then B . Given $PT = 3$ and $PA = 1$, write down PB .
- Write down the length of the tangent from $(7, 0)$ to $x^2 + y^2 = 36$.
- Two tangents from the point P to a circle of unknown radius make an angle of 60° and $OP = 6$. Write down the length of each tangent.
- The two tangents from P to the circle $x^2 + y^2 = 25$ make an angle of 120° . Write down OP .
- In the cyclic quadrilateral $ABCD$, $\angle BAC = 30^\circ$. Write down $\angle BDC$.

Section B Manipulation Drills

(≤ 50 s each)

Draw the radius to the point of contact: it is perpendicular to the tangent. Then look for a right angle or a power of a point.

1. A tangent at T makes an angle of 55° with chord TA . The angle in the alternate segment subtended by TA is?
 - A) 35°
 - B) 55°
 - C) 110°
 - D) 70°
2. $ABCD$ is cyclic with $\angle A = 2x$ and $\angle C = 3x$. Find x .
 - A) 24
 - B) 30
 - C) 36
 - D) 72
3. A *tangential* (circumscribed) quadrilateral $ABCD$ has $AB = 5$, $BC = 7$ and $CD = 6$. By Pitot's theorem ($AB + CD = BC + DA$), find DA .
 - A) 4
 - B) 6
 - C) 8
 - D) 18
4. From external P , tangent $PT = 12$ touches a circle and secant PAB meets it with $PA = 8$. Find the chord AB .
 - A) 6
 - B) 10
 - C) 12
 - D) 18
5. From an external point P , two secants meet the same circle at $P-A-B$ and $P-C-D$, with $PA = 4$, $PB = 18$ and $PC = 6$. Write down the length of the chord CD .
6. The point P has $OP = 6$ with respect to a circle of radius 3. The angle between the two tangents from P is?
 - A) 30°
 - B) 45°
 - C) 60°
 - D) 90°
7. Two circles of radius 6 each pass through the other's centre. They meet at A and B . How long is AB ? (after *Primer 1.4 P16*)
 - A) 6
 - B) 12



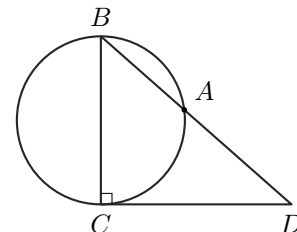
- C) $6\sqrt{2}$
 D) $6\sqrt{3}$
 E) $3\sqrt{3}$
8. Two concentric circles have radii in the ratio $2 : 5$. AC is a diameter of the larger circle, and the chord BC of the larger circle touches the smaller circle. If $AB = 8$, what is the radius of the larger circle? (after Primer 1.4 P13)
- A) 16
 B) 20
 C) 8
 D) 12
 E) 10
9. The line ST is a chord of the larger of two concentric circles and touches the smaller one. If $ST = 30$, find the area of the region between the circles. (after Primer 1.4 P8)
10. AB is a diameter of a circle and C is a point on the circle with $\angle ABC = 55^\circ$. Write down $\angle BAC$.

Section C Substitution & Structure

(≤ 90 s each)

SMC Q18–22 level. Join the centres; tangent circles have centres $r_1 + r_2$ apart and a common point on that line.

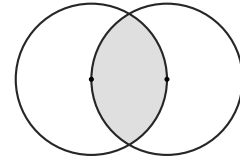
1. BC is a diameter of a circle and CD is the tangent at C . The line BD meets the circle again at A . If $AB = 7$ and $AD = 9$, what is the radius of the circle? (after Primer 1.4 P40)
- A) 12
 B) $4\sqrt{7}$
 C) $2\sqrt{7}$
 D) 8
 E) 6



2. A regular octagon is inscribed in a circle of radius 2, and X is any point on the circle. What is the sum of the squares of the distances from X to the eight vertices? (after SMC 2025 Q21)
- A) 64
 B) 32
 C) 128
 D) 48
 E) 16

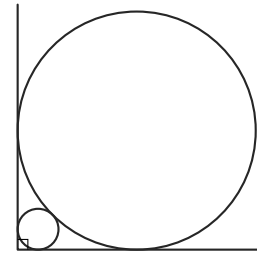
3. Two circles of radius 2 each pass through the other's centre. What is the area of the shaded region common to both circles? (after *Primer 1.4 P24*)

- A) $\frac{4\pi}{3} - \sqrt{3}$
 B) $\frac{8\pi}{3} - \sqrt{3}$
 C) $\frac{4\pi}{3} + 2\sqrt{3}$
 D) $\frac{8\pi}{3} - 2\sqrt{3}$
 E) $2\pi - 2\sqrt{3}$



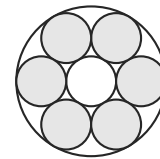
4. A circle of radius 1 touches both arms of a right angle. A larger circle touches both arms and the first circle. What is the radius of the larger circle? (after *Primer 1.4 P35*)

- A) $1 + \sqrt{2}$
 B) $2 + \sqrt{2}$
 C) $2\sqrt{2}$
 D) 3
 E) $3 + 2\sqrt{2}$



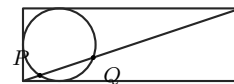
5. Six equal discs lie between two concentric circles. Each disc touches its two neighbours and both circles. The inner circle has radius 1. What is the radius of the outer circle? (after *SMC 2016 Q21*)

- A) 3
 B) $2 + \sqrt{3}$
 C) $1 + \sqrt{3}$
 D) $2\sqrt{3}$
 E) 4



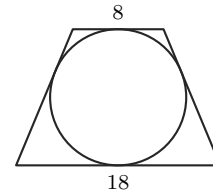
6. A circle of radius 1 touches three sides of a 6×2 rectangle. A diagonal of the rectangle cuts the circle at P and Q . What is the length PQ ? (after *SMC 2017 Q19*)

- A) $\frac{4\sqrt{5}}{5}$
 B) $\frac{2\sqrt{15}}{5}$
 C) $\sqrt{3}$
 D) $\frac{6}{5}$
 E) $\frac{2\sqrt{10}}{5}$



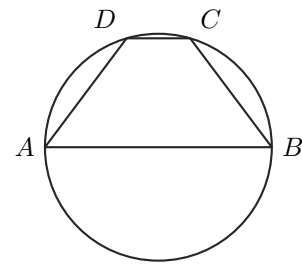
7. An isosceles trapezium has parallel sides of lengths 8 and 18, and all four of its sides touch a circle. What is the radius of the circle? (after SMC 2012 Q20)

- A) $\frac{13}{2}$
 B) 5
 C) 12
 D) 6
 E) $4\sqrt{3}$



8. $ABCD$ is a cyclic quadrilateral in which AB is a diameter of the circle, $AB = 25$ and $AD = BC = 15$. What is the length CD ? (after Primer 1.4 P33)

- A) 10
 B) 15
 C) 13
 D) 9
 E) 7



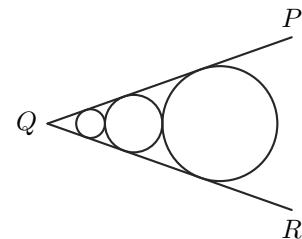
Section D Challenge

(SMC Q23–25 / BMO1 difficulty)

Coordinates, the sine rule or a chain of similar figures. Diagrams are not to scale.

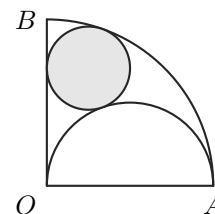
1. Three circles each touch both arms QP and QR of an angle, and each touches the next. The largest radius is 4 times the smallest. What is $\cos \angle PQR$? (after SMC 2011 Q24)

- A) $\frac{7}{9}$
 B) $\frac{1}{3}$
 C) $\frac{2}{3}$
 D) $\frac{8}{9}$
 E) $\frac{4\sqrt{2}}{9}$



2. OAB is a quarter circle with radius 4. A semicircle is drawn on OA as diameter, inside the quarter circle. A circle touches OB , the semicircle and the arc AB . What is its radius? (after SMC 2014 Q19)

- A) $\frac{4}{3}$
 B) 1
 C) $\frac{2}{3}$
 D) $\sqrt{2}$
 E) $\frac{3}{4}$



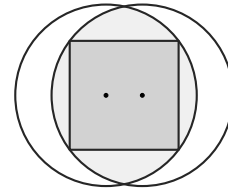
3. A triangle with angles 30° , 45° and 105° is inscribed in a circle of radius 2. What is its area?

(after SMC 2021 Q22)

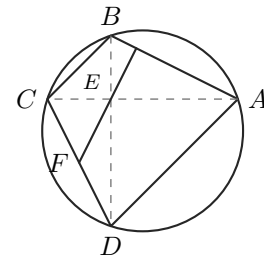
- A) $2 + \sqrt{3}$
- B) $2\sqrt{3}$
- C) $1 + \sqrt{3}$
- D) $3 + \sqrt{3}$
- E) 2

4. Two circles of radius 5 have centres 2 apart. A square is inscribed in the region common to both circles, with its sides parallel and perpendicular to the line of centres. What is the area of the square? (after SMC 2019 Q25)

- A) 64
- B) 32
- C) 25
- D) 36
- E) 48



5. $ABCD$ is a cyclic quadrilateral whose diagonals are perpendicular and meet at E . The line through E perpendicular to AB meets CD at F . Prove that F is the midpoint of CD .



“With circles, draw what you cannot see: the centres, the radii to the points of contact, and the line joining two centres.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths